

quantum dot

The equivalent of a binary on/off switch in a quantum computer. Unlike switches in the everyday world, a quantum dot can exist in a superposition of states.

quantum electrodynamics (QED)

The quantum theory of how charged particles and photons interact with each other.

quantum leap

The smallest change it is possible for a system to make. Such change involves a random choice from a set of possible changes.

quantum mechanics

The description of the workings of the world on very small scales in terms of the laws of quantum physics.

quantum physics

The laws of physics that apply on very small scales (typically, the scale of atoms and below) where change does not occur smoothly but in discrete steps, or quanta.

quark

Fundamental particle at a level below that of the proton and the neutron. There are three quarks in every proton, and three quarks in every neutron.

radioactive decay

The process whereby an unstable particle or atomic nucleus transforms itself into a different particle (or nucleus), usually by spitting out another particle,

such as an electron or an alpha particle.

radiation

Any kind of energy traveling through space. Light is a form of electromagnetic radiation.

Schrödinger equation

A wave equation that describes the behavior of electrons (and other "particles") in quantum mechanical terms.

Strong nuclear force

Holds protons and neutrons together to make nuclei.

superposition of states

State of a quantum system when its wave function is made up of a mixture of waves corresponding to different physical possibilities.

thought experiment

An imaginary experiment, only carried out "in the mind," designed to demonstrate logically some point of view. Schrödinger's Cat Paradox is a classic example.

ultraviolet catastrophe

Prediction from classical mechanics that a hot object should radiate vast amounts of energy at short wavelengths, whatever its temperature. This is a catastrophe for classical theory, pointing the way to quantum physics, because it does not match the way black bodies radiate.

virtual particles

The particles exchanged between "real" particles during interactions—for example, the photons that carry the electromagnetic interaction between two electrons. They only exist while carrying the force.

w particles

Two of the intermediate vector bosons (*see also* z particle).

wave function

Equation describing the properties of a quantum entity, such as an electron, in terms of waves.

wave mechanics

A version of quantum mechanics based on the wave equation discovered by Erwin Schrödinger.

weak interaction

The "force" responsible for some kinds of radioactive decay.

x-rays

Electromagnetic radiation with wavelengths in the range from 12×10^{-9} to 12×10^{-12} m, slightly longer than gamma rays.

Young's slit experiment

Another name for the double-slit experiment, after Thomas Young, who carried out the experiment with light at the end of the 1700s.

z particle

One of the intermediate vector bosons (*see also* w particles).

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